

# Awareness of, and participation with, digital alcohol marketing, and the association with frequency of high episodic drinking among young adults

Nathan Critchlow, Crawford Moodie, Linda Bauld, Adrian Bonner & Gerard Hastings

**To cite this article:** Nathan Critchlow, Crawford Moodie, Linda Bauld, Adrian Bonner & Gerard Hastings (2016) Awareness of, and participation with, digital alcohol marketing, and the association with frequency of high episodic drinking among young adults, *Drugs: Education, Prevention and Policy*, 23:4, 328-336, DOI: [10.3109/09687637.2015.1119247](https://doi.org/10.3109/09687637.2015.1119247)

**To link to this article:** <https://doi.org/10.3109/09687637.2015.1119247>



Published online: 28 Jan 2016.



Submit your article to this journal [↗](#)



Article views: 837



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 3 View citing articles [↗](#)



## Awareness of, and participation with, digital alcohol marketing, and the association with frequency of high episodic drinking among young adults

Nathan Critchlow<sup>1</sup>, Crawford Moodie<sup>1</sup>, Linda Bauld<sup>1</sup>, Adrian Bonner<sup>2</sup>, and Gerard Hastings<sup>1</sup>

<sup>1</sup>Institute for Social Marketing, University of Stirling, Stirling, UK and <sup>2</sup>School of Applied Social Sciences, University of Stirling, Stirling, UK

### Abstract

**Aim:** To explore the association between awareness of traditional and digital marketing, participation with digital marketing and young adults' frequency of high episodic drinking (HED). **Methods:** An online cross-sectional survey of 18–25 year olds ( $n = 405$ ) measured awareness of nine traditional marketing channels, and awareness of, and participation with, 11 digital marketing channels. HED was measured using the final item from the Alcohol Use Disorders Identification Test – Consumption (AUDIT-C). **Findings:** Respondents, on average, were aware of alcohol being marketed through 4.30 traditional and 6.23 digital marketing channels, and had participated with marketing through 2.34 digital channels. Respondents who reported HED on at least a weekly basis reported the most awareness of, and participation with, alcohol marketing. Those who reported never engaging in HED, or doing so less than monthly, reported the lowest. Significant associations were found between awareness of, and participation with, traditional and digital alcohol marketing and increased frequency of HED. **Conclusions:** That digital marketing was more successful than traditional in reaching young adults, and had a stronger association with increased frequency of HED, highlights the dynamic nature of marketing communications and the need for further research to fully understand young people's experience with digital marketing.

### Keywords

Alcohol marketing, alcohol, binge drinking, digital alcohol marketing, high episodic drinking, marketing, quantitative research, traditional alcohol marketing, young adults

### History

Received 5 September 2015

Revised 30 October 2015

Accepted 9 November 2015

Published online 21 December 2015

### Introduction

Globally, the prevalence of high episodic drinking (HED) is highest among young people. It is particularly high in Europe, where the proportion of 15–19 year olds who report HED (31.2%) is almost double that of the rest of the adult population (16.5%) (World Health Organisation, 2014). This is problematic as excessive alcohol consumption is linked with increased mortality (Rehm & Sheild, 2013), morbidity (Shield, Parry, & Rehm, 2013) and a range of social harms (World Health Organisation, 2014), with the economic burden substantial. In England, the accumulated social and economic costs of alcohol are estimated to be £21 billion (Public Health England, 2014), a figure which rises to €125 billion in the European Union (Anderson & Baumberg, 2006) and \$233 billion in the United States (Bouchery, Harwood, Sacks, Simon, & Brewer, 2011).

Reviews of research exploring the impact of alcohol marketing suggest a causal relationship between young peoples' exposure to alcohol marketing and alcohol use (Anderson, de Bruijn, Angus, Gordon, & Hastings, 2009; Gordon, Hastings, & Moodie, 2010a; Smith & Foxcroft, 2009). However, further research is required given the alcohol industry's ability to innovate marketing practice

(Jernigan, 2009), for instance via digital media. Since the turn of the century, there has been increased commercial interest in how the internet can be used to reach consumers. Indeed, in the last six years alone commercial expenditure on digital advertising in the UK has increased from £3.6 billion to an estimated £7.2 billion in 2014, a figure which represents 39% of all advertising expenditure (OFCOM, 2015a). Econometric data, however, does not account for additional below-the-line forms of digital marketing with limited costs, such as branded social media accounts (Brooks, 2010), or how user-generated content further contributes to marketing initiatives (Arnhold, 2010). As such advertising expenditure provides only a snapshot of online marketing activity. Understanding how digital marketing affects young adults' health is important given that the proportion of 16–24 year olds that use the internet is higher than any other age group in the adult population (Office for National Statistics, 2015a). Frequency of internet use and amount of time spent online is also negatively correlated with age (Office for National Statistics, 2015a), and unsurprisingly it is young people who have the highest rates of participation in activities such as social networking, watching on-demand television, gaming, sharing media, shopping and downloading smartphone applications (OFCOM, 2014, 2015b).

The alcohol industry has embraced digital marketing (Baker, 2013; Barnett, 2012; MacLeod, 2013a), evidenced

Correspondence: Nathan Critchlow, Institute for Social Marketing, University of Stirling, RG Bomont Building, Stirling FK9 4LA, UK. E-mail: [nathan.critchlow@stir.ac.uk](mailto:nathan.critchlow@stir.ac.uk)

by the continued redistribution of marketing expenditure to digital channels (Mullman, 2009; Schultz, 2014, 2015). The industry also seems to be reaping the rewards of such investment, with Diageo reporting that sales had increased by 20% as a result of a strategic partnership with Facebook (Diageo PLC, 2011). It has been suggested, however, that the alcohol industry is capitalising on digital marketing to reach young people and influence online and drinking behaviour (Alcohol Concern, 2013; Atkinson, Ross, Begley, & Sumnall, 2014; Barry et al., 2015a; Brooks, 2010; Carah, Brodmerkel, & Shaul, 2015; Gordon, 2011; Griffiths & Casswell, 2010; Purves, Stead, & Eadie, 2014; Siegel et al., 2015; Winpenny, Marteau, & Nolte, 2013). In particular, the ability to publish and relate marketing to real-time events (Carah, Brodmerkel, & Hernandez, 2014; Nicholls, 2012), interactive “advergaming” (Chester, Montgomery, & Dorfman, 2010), viral campaigns (Dobson, 2012), blending marketing seamlessly with user-created content (Brodmerkel & Carah, 2013), and the ability to submerge marketing below explicit levels of awareness (Lyons et al., 2014), have all served to make digital marketing more powerful and less controllable than traditional marketing (Hastings & Sheron, 2013). This creates unique challenges for researchers and regulators alike (Brodmerkel & Carah, 2013; Brooks, 2010; Carah et al., 2015; Carroll & Donovan, 2002; Chester et al., 2010; European Centre for Monitoring Alcohol Marketing, 2009; Jernigan & Rushman, 2014; McCreanor et al., 2013; Nicholls, 2012).

Self-report research has found awareness of digital alcohol marketing in young people to range from 5% for e-mail marketing (Gordon, Harris, Mackintosh, & Moodie, 2011) to 66% for display advertisements on websites (de Bruijn, 2013). Evidence further suggests young people are willing to participate, with the proportion doing so ranging from 4% for accessing websites (Gordon et al., 2011) to 32% interacting with marketing on social networking sites (de Bruijn, 2013). Young people’s exposure also appears to be increasing. For instance, a longitudinal study in Scotland reported the proportion of adolescents who were aware of, and participating with, marketing on social networking sites had doubled by two-year follow-up (Gordon, Mackintosh, & Moodie, 2010b). Evidence further suggests exposure to digital marketing has an association with alcohol use and HED (de Bruijn, 2013; Gordon et al., 2011; Jones & Magee, 2011; Lin et al., 2012), a relationship reported as causal by two longitudinal studies (de Bruijn, Tanghe, & Beccaria, 2012; Gordon et al., 2010b). It has also been suggested that participation with digital alcohol marketing has a stronger association with alcohol use, despite greater awareness and participation with traditional alcohol marketing (Gordon et al., 2011; Lin et al., 2012).

In this study, we build upon past research by exploring the association between digital alcohol marketing and frequency of HED among young adults (18–25 years old). This age group represent a potentially vulnerable population given they report some of the highest rates of internet use and HED drinking in the UK (Office for National Statistics, 2015a, 2015b). They also represent a key target market for the alcohol industry (Hastings et al., 2010) and have been overlooked in past research. We also consider both awareness of, and participation with, a wider range of digital marketing

channels than has been used in past research, and also awareness of alcohol marketing in traditional (non-digital) channels to provide comparison.

## Methods

### Design and sample

An online, cross-sectional survey was conducted with 18–25 years old in the UK ( $n = 405$ ). Responses were collected in two six-month waves, November 2013–April 2014 and February 2015–July 2015. A convenience sample was recruited through adverts posted in three online contexts. First, adverts were posted on the “Surveys and Research” section of [www.thestudentroom.co.uk](http://www.thestudentroom.co.uk), a large UK-based community website for school and university students. Use of internet forums for recruitment in addiction and substance use literature is increasing (Norman, Grace, & Lloyd, 2014; Pontes, Szabo, & Griffiths, 2015), with the website used in this study having been used in previous research (Corazza, Simonato, Cokery, Trincas, & Schifano, 2014; Furnell, Bryant, & Phippen, 2007). Second, an advert was placed on an internal university website, which allowed first- and second-year undergraduate psychology students to partake in the study in exchange for academic credits which were required for a research methods module. Eligible students had not covered content related to the aims of the study, or marketing in general, as part of their undergraduate degree course. Third, an advert was posted on social networking site Facebook, a method of convenience sampling previously used in alcohol marketing research (Jones & Magee, 2011).

## Measures

### Demographics

Age, gender, ethnicity and religious denomination were measured.

### Internet use

To measure frequency of internet use respondents were asked “How often do you use the internet?” Responses were given on a nine-point scale (1 = never; 9 = everyday). To measure duration of internet use respondents were asked “On average, how long do you spend on the internet per day?” Responses were given on a nine-point scale (1 = I don’t use the internet; 9 = nine or more hours).

### Awareness of traditional (offline) alcohol marketing

Respondents were asked “Think about ways you have seen alcohol marketed in the last 30 days. How was it marketed?” They were presented with a list of nine traditional (offline) marketing channels used in previous exposure research (Gordon et al., 2011; Lin et al., 2012); (i) event sponsorship, (ii) magazines, (iii) television, including programme sponsorship, (iv) sports sponsorship, (v) in-store posters, (vi) billboards and posters, (vii) newspapers, (viii) packaging, and (ix) price promotions. Respondents indicated all those they had seen alcohol marketed through by clicking a “Yes” response icon next to each channel. Responses were summed to create a total awareness score (0–9).

**Awareness of, and participation with, digital marketing**

Awareness of, and participation with, digital marketing was assessed for 11 channels; (i) downloadable content for mobile phones and computers (e.g. screensavers), (ii) adverts on on-demand television, (iii) games, (iv) online shops selling alcohol or branded merchandise, (v) websites, (vi) marketing on social media (Facebook and Twitter), (vii) viral videos or adverts, (viii) smartphone applications, (ix) display advertisements, (x) e-mails, and (xi) competitions. For each, respondents were shown a visual example and asked to indicate if they had seen alcohol marketed through that channel (Yes/No), and whether they had participated with marketing through that channel (Yes/No). The phrasing of each question was specific to the channel and the keywords emphasised in bold to ensure clarity over what constituted awareness and participation, as this differed across channels (Figure 1). For each question respondents were reminded that the visual was only an example and to consider others they recalled. Responses were summed to create a total score for both awareness and participation (0–11).

**Frequency of high episodic drinking**

Frequency of HED was measured using the final item from the Alcohol Use Disorders Identification Test – Consumption (AUDIT-C), a three-item screening instrument recently validated as an appropriate alcohol screening tool in the UK (Foxcroft, Smith, Thomas, & Howcutt, 2015), used in research with young adults (Barry, Chaney, Stellefson, & Dodd, 2015b; Davoren, Shiely, Byrne, & Perry, 2015) and considered an appropriate brief screening tool by the National Health Service (National Institute for Health and Care Excellence, 2010). Respondents were asked “How often have you had six or more units if female, or eight or more units if male, on a single occasion in the last year?” Responses were given on a five-item scale: 0, never; 1, less than monthly; 2, monthly; 3, weekly; and 4, daily or almost daily. A diagram depicting the unit content of various common alcoholic drinks, such as a 175 ml glass of wine or pint of regular beer/lager/cider, was included to assist in

estimating the units consumed on previous drinking occasions.

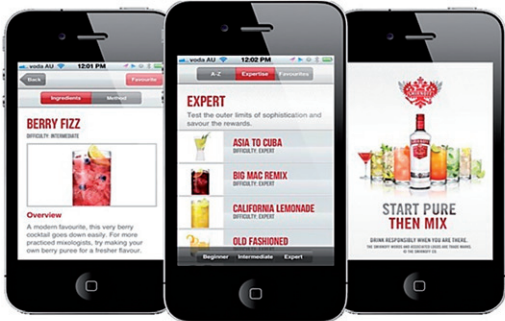
**Procedure**

The self-report questionnaire was designed and hosted on Qualtrics (www.qualtrics.com), a full-featured, web-based tool for creating and conducting online surveys. Advertisements highlighting the aims and methods of the study were placed on the “Surveys and Research” section of www.thestudentroom.co.uk. The adverts remained there throughout the two recruitment periods and were refreshed once a fortnight. An advert was also uploaded to the internal student recruitment website and posted on social networking site, Facebook, by the lead investigator. Those who met the inclusion criteria (18–25 years old and living in the UK) were invited to click on a hyperlink at the bottom of the advert, which directed their web-browser to the online questionnaire. Prior to survey onset respondents were informed of the study aims, incentive, the anonymity of data, and the voluntary nature of the study and ability to withdraw at any point. At the bottom of the first page respondents were asked to confirm they met the age criteria and they consented to take part (Yes/No). The average completion time for the questionnaire was 20 minutes. After the questionnaire had been completed, a debriefing page was displayed that asked respondents whether they would like to enter themselves into the ballot to win one-of-five £10 vouchers for an e-retailer (Yes/No) as a thank you for participating in the research. Those who wished to enter were redirected to a separate online questionnaire to provide an e-mail address which registered their place. Those who did not were directed to final page that confirmed the questionnaire was finished. Ethical approval was obtained from the University of Roehampton’s ethics board (PSYC 13/097) and the Salvation Army’s Territorial Health Ethics Advisory Committee (4 October 2013).

**Statistical analysis**

Data were analysed using SPSS version 21 (SPSS Inc., Chicago, IL). Cases in which (i) no age was specified, (ii) the respondent was not aged 18–25 years old, and (iii) the IP

Figure 1. Example of grid for measuring awareness of, and participation with, digital alcohol marketing.  
Image credit (Diageo PLC, 2012).



**Smartphone applications or mobile games made by an alcohol company (e.g. Carling's 'iPint' application or Smirnoff's 'Recipe App').**

|                                                                                                 | Yes                   | No                    |
|-------------------------------------------------------------------------------------------------|-----------------------|-----------------------|
| Have you <b>seen</b> a smartphone application or mobile game created by an alcohol company?     | <input type="radio"/> | <input type="radio"/> |
| Have you <b>downloaded, used or played</b> a smartphone application made by an alcohol company? | <input type="radio"/> | <input type="radio"/> |



address of the respondent was outside the UK, were removed. All responses were screened for contradictory answers using cross-tabulation tables. Responses in which participation with digital marketing was indicated, yet awareness of the same channel was not, were removed. This left a sample of 405. Frequency tables were produced to explore awareness of each traditional and digital marketing channel, and participation with each digital channel. A paired samples *t*-test was conducted to examine the differences between cumulative awareness of, and participation with, digital marketing. Three one-way analyses of variance (ANOVAs), each with 10 polynomial linear planned contrasts, were conducted to explore whether awareness of traditional and digital marketing, and participation with digital marketing, differed across the five frequency categories of HED; (i) never, (ii) less than monthly, (iii) monthly, (iv) weekly, (v) daily or almost daily.

A hierarchical linear regression, based on 2000 bootstrapped samples, was conducted with awareness of, and participation with, alcohol marketing as the independent variables and frequency of HED across the sample the dependent variable (0 = never, 4 = daily or almost daily). Demographic variables of age, gender, religious beliefs and ethnicity were entered at step one. Three of these demographic variables were categorical. Male was used as the baseline for gender. Ethnicity was coded into binary categories, ‘‘White British’’ and ‘‘Other’’, and the latter was used as the baseline. Religion was also coded into binary categories based on whether respondent was religious (Yes/No). ‘‘No’’ was used as the baseline. Cumulative awareness of traditional marketing was entered at step two, cumulative awareness of digital marketing was entered at step three, and cumulative participation with digital marketing at step four. The marketing variables were entered in this order to; (i) compare the association awareness of traditional marketing had with increased frequency of HED to the association that awareness of digital marketing had with increased frequency of HED, and (ii) to compare both forms of awareness against the association that participation with digital marketing had with increased frequency of HED. All variables were entered using a forced entry method.

## Results

### Sample characteristics

The sample ( $n = 405$ ) had an average age of 21 ( $SD = 2.22$ ). There were 113 males (28%) and 292 females (72%). Most respondents identified as White British (66%), with the rest 11% Asian, 5% European, 4% Caribbean, 3% African, 3% from other ethnicities, 3% Middle Eastern, 2% mixed ethnicity, 2% who did not wish to state ethnicity, >1% North American and >1% South American/Hispanic. The majority (62%) of the sample indicated they were not religious. Almost the entire sample used the internet everyday (99%), spending an average of three–four hours online.

### Awareness of traditional marketing

Respondents, on average, reported awareness of alcohol being marketed through 4.30 ( $SD = 2.29$ ) of the nine traditional channels, with a range of 32–80% (Table 1). Those who

reported HED weekly also reported the most awareness, whilst those who reported never having engaged in HED reported the least (Table 3). A one-way ANOVA indicated awareness of traditional marketing differed significantly across the five HED categories, Welch’s  $F(4, 51.92) = 3.15$ ,  $p < 0.05$ ,  $\eta^2 = 0.02$ . *Post-hoc* planned contrasts, in which equal variances were not assumed, revealed those who reported never engaging in HED reported significantly lower awareness of traditional marketing than those who reported HED monthly,  $t(202.96) = -2.73$ ,  $p < 0.01$ ,  $d = 0.38$ ; or weekly,  $t(91.66) = -3.24$ ,  $p < 0.005$ ,  $d = 0.68$ . Those who reported HED weekly also reported significantly more awareness than those who reported HED monthly,  $t(81.61) = -2.05$ ,  $p < 0.05$ ,  $d = 0.45$ . The remaining *post-hoc* contrasts were non-significant.

### Awareness of digital alcohol marketing

Respondents, on average, reported awareness of 6.23 ( $SD = 2.52$ ) of the 11 digital marketing channels, with a range of 29–88% (Table 2). Those who reported HED weekly reported the most awareness, whilst those who reported never engaging in HED reported the least (Table 3). A one-way

Table 1. Awareness of alcohol marketing through traditional (offline) channels.

| Channel                                      | %           | <i>n</i> |
|----------------------------------------------|-------------|----------|
| Television (including programme sponsorship) | 80          | 323      |
| Price promotion                              | 65          | 262      |
| In-store posters                             | 54          | 217      |
| Billboards/posters                           | 54          | 219      |
| Magazines                                    | 42          | 168      |
| Packaging                                    | 35          | 143      |
| Event sponsorship                            | 35          | 143      |
| Newspapers                                   | 35          | 141      |
| Sports sponsorship                           | 32          | 127      |
| Average (%)                                  | 48          | 193      |
| Cumulative mean ( <i>SD</i> )                | 4.30 (2.29) |          |

Table 2. Awareness of, and participation with, digital alcohol marketing.

| Channel                        | Awareness   |          | Participation |          | Change |
|--------------------------------|-------------|----------|---------------|----------|--------|
|                                | %           | <i>n</i> | %             | <i>n</i> |        |
| On-demand TV ads               | 88          | 355      | 49            | 199      | –39    |
| Videos                         | 84          | 340      | 54            | 220      | –30    |
| Display advertisements         | 76          | 309      | 12            | 49       | –64    |
| Social media                   | 65          | 262      | 20            | 81       | –45    |
| Online shop                    | 62          | 249      | 15            | 61       | –47    |
| Competitions                   | 57          | 229      | 16            | 66       | –41    |
| Websites                       | 50          | 202      | 28            | 114      | –22    |
| Smartphone applications        | 46          | 186      | 20            | 79       | –26    |
| Games                          | 35          | 141      | 10            | 39       | –25    |
| E-mail                         | 33          | 133      | 6             | 26       | –27    |
| Downloadable content           | 29          | 116      | 3             | 11       | –26    |
| Average (%)                    | 57          | 229      | 21            | 86       | –36    |
| Cumulative mean ( <i>SD</i> )* | 6.23 (2.52) |          | 2.34 (1.96)   |          | 3.89   |

Cumulative mean refers to the average number of the 11 digital marketing channels, each participant was aware of, and participating with.

\*Paired samples *t*-test: awareness versus participation,  $t(404) = 39.22$ ,  $p < 0.001$ .

ANOVA indicated awareness of digital marketing differed significantly across the five HED categories,  $F(4, 400) = 4.18$ ,  $p < 0.005$ ,  $\eta^2 = 0.04$ . *Post-hoc* planned contrasts, in which equal variances were assumed, indicated those who reported never engaging in HED reported significantly lower awareness of digital marketing than those who reported HED less than monthly,  $t(400) = 1.93$ ,  $p < 0.05$ ,  $d = 0.19$ ; monthly,  $t(400) = -3.61$ ,  $p < 0.001$ ,  $d = 0.36$ ; or weekly,  $t(400) = -3.01$ ,  $p < 0.005$ ,  $d = 0.30$ . Similarly those who reported HED less than monthly reported less awareness than those who reported HED monthly,  $t(400) = -2.01$ ,  $p < 0.05$ ,  $d = 0.20$ .

### Participation with digital alcohol marketing

Respondents, on average, reported participation with 2.34 ( $SD = 1.96$ ) of the 11 digital marketing channels, with a range of 3–54% (Table 2). A one-way ANOVA indicated participation with digital marketing differed significantly across the five HED groups, Welch's  $F(4, 50.61) = 12.45$ ,  $p < 0.001$ ,  $\eta^2 = 0.10$ . There was also a significant linear trend,  $F(1, 400) = 16.47$ ,  $p < 0.001$ , indicating as frequency of HED increased, participation also increased (Table 3). *Post-hoc* planned contrasts, in which equal variances were not assumed, revealed those who reported never engaging in HED reported significantly less participation than those who reported HED less than monthly,  $t(241.51) = -5.05$ ,  $p < 0.001$ ,  $d = 0.65$ ; monthly,  $t(193.185) = -5.63$ ,  $p < 0.001$ ,  $d = 0.81$ ; weekly,  $t(53.22) = -4.59$ ,  $p < 0.001$ ,  $d = 1.26$ ; and daily or almost daily,  $t(8.33) = -2.40$ ,  $p < 0.05$ ,  $d = 1.66$ .

### Association with frequency of high episodic drinking

Twenty-four per cent ( $n = 97$ ) of the sample had never engaged in HED, 37% ( $n = 151$ ) did so less than monthly, 27% ( $n = 108$ ) did so monthly, 10% ( $n = 40$ ) did so weekly and 2% ( $n = 9$ ) did so daily or almost daily. The first step of the linear regression model examined the association between frequency of HED and the demographic variables (Table 4). This step was statistically significant,  $F(4, 400) = 8.65$ ,  $p < 0.001$ , and accounted for 8% of the variance in frequency of HED. Within this step being male, of older age, and being of White British ethnicity (as opposed to from other) was associated with more frequent HED. Religion had no significant association with frequency of HED.

The second step of the model examined the association between frequency of HED and awareness of traditional marketing. This step was statistically significant,

$F(1, 399) = 7.62$ ,  $p < 0.01$ , and increased the variance of HED explained to 10% ( $\Delta R^2 = 0.017$ ). This indicated the more traditional marketing channels a respondent was aware of the more likely they were to engage in HED more frequently.

The third step of the model examined the association between frequency of HED and awareness of digital marketing. This step was also statistically significant,  $F(1, 398) = 4.66$ ,  $p < 0.05$ , and increased the total variance of HED explained to 11% ( $\Delta R^2 = 0.010$ ). This indicated the more digital marketing channels a respondent was aware of the more likely they were to report HED more frequently. In step three, awareness of traditional marketing ceased to be a significant predictor when adjusting for awareness of digital marketing ( $\beta = 0.09$ ,  $p = 0.09$ ). This indicated the association between increased awareness of digital marketing and increased frequency of HED was stronger than the association between increased awareness of traditional marketing.

The fourth step of the regression model examined the association between frequency of HED and participation with digital marketing. This step was also significant,  $F(1, 397) = 13.43$ ,  $p < 0.001$ , and increased the variance of HED explained to 14% ( $\Delta R^2 = 0.029$ ). This indicated the more digital marketing channels a respondent was participating with the more likely they were to report HED more frequently. In step four, awareness of both traditional ( $\beta = 0.08$ ,  $p = 0.10$ ) and digital marketing ( $\beta = -0.02$ ,  $p = 0.73$ ) ceased to be significant predictors when adjusting for participating with digital marketing. This indicated the association between increased participation with digital marketing and increased frequency of HED was stronger than increased awareness of either digital or traditional marketing.

### Discussion

The results are consistent with past research, by demonstrating a significant association between awareness of, and participation with, alcohol marketing and increased frequency of HED. The results extend knowledge given that research exploring digital marketing has only considered limited channels (de Bruijn, 2013; de Bruijn et al., 2012; Gordon et al., 2010b, 2011; Jones & Magee, 2011; Lin et al., 2012). Young adults are increasingly using the internet for recreational activities such as, gaming, sharing media, shopping and downloading smartphone applications (OFCOM, 2015b).

Table 3. One-way analysis of variance, awareness and participation with alcohol marketing by frequency of high episodic drinking.

| Form of exposure                | Frequency of high episodic drinking |           |                                        |           |                              |           |                            |           |                                          |           | <i>F</i>           | <i>p</i> |
|---------------------------------|-------------------------------------|-----------|----------------------------------------|-----------|------------------------------|-----------|----------------------------|-----------|------------------------------------------|-----------|--------------------|----------|
|                                 | Never<br>( <i>n</i> = 97)           |           | Less than<br>monthly ( <i>n</i> = 151) |           | Monthly<br>( <i>n</i> = 108) |           | Weekly<br>( <i>n</i> = 40) |           | Daily or almost<br>daily ( <i>n</i> = 9) |           |                    |          |
|                                 | <i>M</i>                            | <i>SD</i> | <i>M</i>                               | <i>SD</i> | <i>M</i>                     | <i>SD</i> | <i>M</i>                   | <i>SD</i> | <i>M</i>                                 | <i>SD</i> |                    |          |
| Awareness traditional marketing | 3.77                                | 2.17      | 4.23                                   | 2.32      | 4.64                         | 2.38      | 4.90                       | 1.71      | 4.67                                     | 3.00      | 2.98 <sup>a</sup>  | <0.05    |
| Awareness digital marketing     | 5.49                                | 2.48      | 6.11                                   | 2.51      | 6.75                         | 2.47      | 6.90                       | 2.25      | 6.67                                     | 3.12      | 4.18               | <0.005   |
| Participation digital marketing | 1.36                                | 1.40      | 2.42                                   | 1.91      | 2.69                         | 1.97      | 3.05                       | 2.15      | 3.78                                     | 2.99      | 12.36 <sup>a</sup> | <0.001   |

<sup>a</sup>Homogeneity of variances not assumed and Welch's *F* value reported. All *post-hoc* planned contrasts for ANOVAs based on equal variances not assumed.

Table 4. Association between awareness of, and participation with, alcohol marketing and frequency of high episodic drinking.

| Variables                                   | Unstandardised coefficients |           | Standardised coefficients | <i>t</i> | <i>p</i> | $\Delta R^2$ |
|---------------------------------------------|-----------------------------|-----------|---------------------------|----------|----------|--------------|
|                                             | <i>b</i>                    | <i>SE</i> | $\beta$                   |          |          |              |
| Step one                                    |                             |           |                           |          |          | 0.08         |
| Age                                         | 0.05                        | 0.02      | 0.10                      | 2.08     | <0.05    |              |
| Gender (1 = male, 2 = female)               | -0.35                       | 0.11      | -0.16                     | -3.26    | <0.005   |              |
| Ethnicity (0 = Other, 1 = White British)    | 0.39                        | 0.11      | 0.18                      | 3.69     | <0.005   |              |
| Religion (0 = No, 1 = Yes)                  | -0.13                       | 0.11      | -0.06                     | -1.24    | n.s      |              |
| Step two                                    |                             |           |                           |          |          | 0.02         |
| Awareness of traditional marketing (0-9)    | 0.06                        | 0.02      | 0.13                      | 2.76     | <0.005   |              |
| Step three                                  |                             |           |                           |          |          | 0.01         |
| Awareness of traditional marketing          | 0.04                        | 0.02      | 0.09                      | 1.71     | n.s      |              |
| Awareness of digital marketing (0-11)       | 0.05                        | 0.02      | 0.12                      | 2.16     | <0.05    |              |
| Step four                                   |                             |           |                           |          |          | 0.03         |
| Awareness of traditional marketing          | 0.04                        | 0.02      | 0.08                      | 1.63     | n.s      |              |
| Awareness of digital marketing              | -0.01                       | 0.03      | -0.02                     | -0.35    | n.s      |              |
| Participation with digital marketing (0-11) | 0.12                        | 0.03      | 0.23                      | 3.67     | <0.001   |              |

Sample ( $n = 405$ ); Durbin Watson: 1.78.

n.s: non-significant.

Step two, three and four controlled for all four demographic variables from stage one.

Dependent variable (DV): how often do you have six or more units if female, or eight or more if male, on a single occasion in the last year? (0 = never; 4 = daily or almost daily).

Unstandardised coefficients, standard errors and *p* values of significance based on 2000 bootstrapped pairwise comparisons.

By exploring these channels in the context of alcohol marketing the findings suggest that the alcohol industry is reactive to changes in how young adults use the internet and have developed digital marketing strategies that appear successful in reaching them and encouraging them to participate.

Compared to previous research, the findings show that average awareness of digital marketing was 16% higher than reported by de Bruijn (2013) and at least 50% higher than reported by Lin et al. (2012) and Gordon et al. (2011). This increase was also true for participation, with the average estimate reported at least 15% higher than adolescents in Scotland or New Zealand (Gordon et al., 2011; Lin et al., 2012). Average awareness of alcohol marketing was also higher for digital than traditional marketing channels. This contrasts with past research in Scotland (Gordon et al., 2011) and New Zealand (Lin et al., 2012) where the average awareness of traditional marketing, which was close to the estimate found in this study, was higher than digital marketing. Therefore the findings suggest that although awareness of traditional marketing has remained stable in recent years, digital marketing has become increasingly successful in reaching young people. It is important to note that as previous research has only focused on adolescents, the findings may simply reflect the greater amount of online marketing young adults are exposed to. It may also be that the inclusion of a visual example for each digital marketing channel stimulated cue-reactivity, which is known facilitate better recall (McCracken & Macklin, 1998). Alternatively, the increase in exposure may simply reflect the greater expenditure on digital marketing (OFCOM, 2015a) or increased internet use by young adults (Office for National Statistics, 2015a). However, regardless of the reason, the increased exposure reported does help to explain the alcohol industry's shift in promotional expenditure towards new media.

The findings are consistent with reviews of research exploring traditional alcohol marketing, which suggest an association between exposure and alcohol use (Anderson et al., 2009; Gordon et al., 2010a; Smith & Foxcroft, 2009), and also support formative research which has suggested this link is also true for digital marketing (de Bruijn et al., 2012; de Bruijn, 2013; Gordon et al., 2011; Lin et al., 2012; Jones & Magee, 2011). Our findings are also consistent with suggestions that digital marketing has resulted in a more powerful association with consumption compared to traditional marketing (Hastings & Sheron, 2013), and that participation has a stronger association than awareness (Gordon et al., 2011; Lin et al., 2012). Evidence suggests that the alcohol industry is also aware of this and is encouraging consumers to engage with digital marketing (Atkinson et al., 2014; Nicholls, 2012), even by adding quick response barcodes and augmented reality features to offline marketing such as product packaging (Purves et al., 2014), beer mats (Spencer, 2014) and pint glasses (Nudd, 2012). Nonetheless, participation with digital marketing was lower than awareness of both traditional and digital marketing. It therefore remains plausible some young adults who may not participate may still be aware of digital and traditional marketing, which the findings suggest is also associated with increased frequency of HED, albeit not as strongly as participation. Past research suggests the association between marketing and consumption is best understood as cumulative across all marketing channels (Gordon et al., 2011; Lin et al., 2012) and our findings further support that all three forms of exposure should be considered in the relationship between exposure and alcohol use in young adults.

This study has a number of limitations. First, the findings are cross-sectional and do not provide evidence that marketing increased the frequency HED over time. The suggestion that marketing does have a causal relationship with

consumption is supported by longitudinal research on traditional (Anderson et al., 2009; Smith & Foxcroft, 2009) and digital marketing (de Bruijn et al., 2012), especially studies which have controlled for multiple confounding variables (Gordon et al., 2010b). A causal hypothesis is also supported by message interpretation research, which suggests exposure to alcohol marketing increases cognitions which in turn influence alcohol use. Examples from research include; expectancies, desirability, drinker identity and brand allegiance (Austin, Chen, & Grube, 2006; Austin & Knaus, 2000; Chen, Grube, Bersamin, Waiters, & Keefe, 2005; Fleming, Thorson, & Aitken, 2004; Grube & Waiters, 2005; McClure, Stoolmiller, Tanski, Engels, & Sargent, 2013). However, only de Bruijn et al. (2012) have provided evidence that these social-cognitive pathways explain the impact of digital marketing on consumption. Further research is required before we can accept, with confidence, that the effect is not attributable to reverse causality or residual confounding factors.

Second, the small and self-selected sample means the findings are not representative of all young people in the UK, and considerably more females chose to participate in the study than males. Indeed, the different modes of recruitment may have also skewed the findings. For instance, that some young adults were recruited via a digital forum and others from a social networking site, and therefore may be more actively engaged in digital activities, provides an alternative hypothesis to explain why awareness of, and participation, with digital marketing was greater than previous research. Indeed, the opt-in process for those in the digital forum and social networking site may also mean that only those with a greater interest in alcohol marketing chose to participate. Furthermore, as remaining participants were recruited from an undergraduate psychology course the possibility of response bias cannot be rejected. This could be the result of familiarity with research methods, or token participation to receive the module credits. Furthermore, as all responses were anonymous it was not possible to compare between the different modes of recruitment. Further research exploring different demographics groups, in particular those under the legal age of consumption, would be welcomed.

Another limitation is the results may underestimate awareness of, and participation with, alcohol marketing and the subsequent association with alcohol use. First, the channels we assessed are not exhaustive and did not consider, for instance, marketing on music streaming services (MacLeod, 2013b). Second, online platforms, in particular social media, enable marketing to be submerged below conscious awareness. This can make it difficult to recognise marketing, which may lead to under-estimations in reported exposure (Lyons et al., 2014). Measurement of awareness and participation only had dichotomous responses. It is possible a “do not know” option may have helped identify when a respondent could not be sure they had been exposed to a marketing channel, but did not feel confident rejecting that they had not. Fourth, it was decided that also measuring participation with traditional marketing may have increased response fatigue and drop-out from the study. We acknowledge, however, there would have been investigative benefit in

exploring participation with both traditional and digital marketing.

Finally, this study did not explore awareness of, and participation with, the user-created content which co-exists with marketing online. This includes user-generated branding, in which users promote their own sense of brand meaning (Arnhold, 2010; Griffiths & Casswell, 2010; Nicholls, 2012), and also user-created alcohol content, in which individuals promote alcohol consumption independent of brand influence (Moreno, Christakis, Egan, Brockman, & Becker, 2012; Morgan, Snelson, & Elison-Bowers, 2010; Ridout, Campbell, & Ellis, 2012).

## Conclusion

In conclusion, this study extends understanding by demonstrating young adults are aware of, and participating with, alcohol marketing through new media channels, which have not been considered in previous research. By demonstrating a significant association between awareness of, and participation with, alcohol marketing and increased frequency of HED, the findings contribute to a growing body of evidence which suggests marketing is associated with alcohol use in young people. It is notable the findings demonstrate this association in young adults, as they are under-researched, report some of the highest estimates of internet use and HED in the UK, and represent a key target market for the alcohol industry. There are limitations which should inform how the findings are interpreted, in particular the high gender skew, cross-sectional design and self-selected sample. Finally, as this study only provides a snapshot of awareness and participation with online media, further studies on explicit and implicit forms of digital marketing and user-created content are still needed to enhance the evidence base.

## Acknowledgements

The authors pay thanks to ALOVE! This group represent The Salvation Army (TSA) for the new generation and are responsible for the implementing the national youth work strategy by delivering worship, discipleship, mission and social action. The authors also thank TSA's Scottish Alcohol and Drug Strategy Group and the Centre for Applied Research and Assessment in Adolescent and Child Wellbeing at the University of Roehampton, particularly Dr. Catherine Gilvarry and Prof. Cecilia Essau for their input in early stages of project development.

## Declaration of interest

This project was supported by a grant supplied by The Salvation Army in the United Kingdom and Republic of Ireland. The authors report no conflict of interest.

## References

- Alcohol Concern. (2013). *New media, new problem? Alcohol, young people and the internet*. London: Alcohol Concern.
- Anderson, P., & Baumberg, B. (2006). *Alcohol in Europe: A public health perspective – A report for the European Commission*. London: Institute for Alcohol Studies.
- Anderson, P., de Bruijn, A., Angus, K., Gordon, R., & Hastings, G. (2009). Impact of alcohol advertising and media exposure on



- adolescent alcohol use: A systematic review of longitudinal studies. *Alcohol and Alcoholism*, 44, 229–243.
- Arnhold, U. (2010). *User generated branding: Integrating user generated content into brand management*. Bremen: Springer Gabler Research.
- Atkinson, A.M., Ross, K.M., Begley, E., & Sumnall, H. (2014). *Constructing alcohol identities: The role of social network sites (SNS) in young peoples' drinking cultures*. London: Alcohol Research UK.
- Austin, E.W., Chen, M.-J., & Grube, J.W. (2006). How does alcohol advertising influence underage drinking? The role of desirability, identification and scepticism. *Journal of Adolescent Health*, 38, 376–384.
- Austin, E.W., & Knaus, C. (2000). Predicting the potential for risky behavior among those “too young” to drink as the result of appealing advertising. *Journal of Health Communication*, 5, 13–27.
- Baker, J. (2013). Diageo CMO Andy Fennell: “All marketing is digital”. *M & M Global*. Retrieved from <http://www.mandmglobal.com/global-accounts/25-03-13/diageo-cmo-andy-fennell-all-marketing-is-digi.aspx>.
- Barnett, M. (2012). AB InBev shifts spend from TV to social. *Marketing Week*. Retrieved from <https://www.marketingweek.com/2012/03/14/ab-inbev-shifts-spend-from-tv-to-social/>.
- Barry, A.E., Johnson, E., Rabre, A., Darville, G., Donovan, K.M., & Efunbumi, O. (2015a). Underage access to online alcohol marketing content: A YouTube case study. *Alcohol and Alcoholism*, 50, 89–94.
- Barry, A.R., Chaney, B.H., Stellefson, M.L., & Dodd, V. (2015b). Evaluating the psychometric properties of the AUDIT-C among college students. *Journal of Substance Use*, 20, 1–5.
- Bouchery, E.E., Harwood, H.J., Sacks, J.J., Simon, C.J., & Brewer, R.D. (2011). Economic costs of excessive alcohol consumption in the US, 2006. *American Journal of Preventative Medicine*, 41, 516–524.
- Brodmerkel, S., & Carah, N. (2013). Alcohol brands on Facebook: The challenges of regulation brands on social media. *Journal of Public Affairs*, 13, 272–281.
- Brooks, O. (2010). “Routes to magic”: The alcoholic beverage industry's use of new media in alcohol marketing. Stirling: Institute for Social Marketing, University of Stirling.
- Carah, N., Brodmerkel, S., & Shaul, M. (2015). *Breaching the code: Alcohol, Facebook and self-regulation*. Canberra: Foundation for Alcohol Research and Education. Retrieved from [https://espace.library.uq.edu.au/view/UQ:360313/UQ360313\\_OA.pdf](https://espace.library.uq.edu.au/view/UQ:360313/UQ360313_OA.pdf).
- Carah, N., Brodmerkel, S., & Hernandez, L. (2014). Brands and sociality: Alcohol branding, drinking culture and Facebook. *Convergence: The Journal of Research into New Media Technologies*, 20, 259–275.
- Carroll, T.E., & Donovan, R.J. (2002). Alcohol marketing on the internet: New challenges for harm reduction. *Drug and Alcohol Review*, 21, 83–91.
- Chen, M.-J., Grube, J.W., Bersamin, M., Waiters, E., & Keefe, D.B. (2005). Alcohol advertising: What makes it attractive to youth? *Journal of Health Communication*, 10, 553–563.
- Chester, J., Montgomery, K., & Dorfman, L. (2010). *Alcohol marketing in the digital age*. Berkley, CA: Berkley Media Studies Group.
- Corazza, O., Simonato, P., Cokery, J., Trincas, G., & Schifano, F. (2014). “Legal highs: Safe and legal heavens?” A study on the diffusion, knowledge and risk awareness of novel psychoactive drugs among students in the UK. *Revista di Psichiatria*, 49, 89–94.
- Davoren, M.P., Shiely, F., Byrne, M., & Perry, I.J. (2015). Hazardous alcohol consumption among university students in Ireland: A cross-sectional study. *BMJ Open Access*, 5, e006045.
- de Bruijn, A., Tanghe, J., Beccaria, F. (2012). *Report on the impact of European alcohol marketing exposure on youth alcohol expectancies and youth drinking (Deliverable 2.3 and 3.7, Work Package 4)*. Europe: AMPHORA Project.
- de Bruijn, A. (2013). Exposure to online alcohol marketing and adolescents' binge drinking: A cross-sectional study in four European countries. In Anderson, P., Braddick, F., Reynolds, J., & Gual, A. (Eds.), *Alcohol policy in Europe: Evidence from AMPHORA* (2nd ed.) (pp. 63–72). Europe: AMPHORA Project.
- Diageo. (2011). Diageo takes multi-million dollar global partnership with Facebook to the next level [press release]. Retrieved from <http://www.diageo.com/en-row/newsmedia/pages/resource.aspx?resourceid=1072>.
- Diageo. (2012). Smirnoff cocktails for Apple iOS (Version 2.0) [Mobile application software]. Retrieved from <https://itunes.apple.com/gb/app/smirnoff-cocktails/id470110270?mt=8>.
- Dobson, C. (2012). *Alcohol marketing and young people: Time for a new policy agenda*. Kingston: Australian Medical Association.
- European Centre for Monitoring Alcohol Marketing. (2009). *Alcohol Advertising in new media: Trends in alcohol marketing*. European Centre for Monitoring Alcohol Marketing. Retrieved from <http://eucam.info/2009/11/04/alcohol-advertising-in-new-media-trends-in-alcohol-marketing/>.
- Fleming, K., Thorson, E., & Aitken, C.K. (2004). Alcohol advertising exposure and perceptions: Links with alcohol expectancies and intentions to drink or drinking in under-aged youth and young adults. *Journal of Health Communication*, 9, 3–29.
- Foxcroft, D.R., Smith, L., Thomas, H., & Howcutt, S. (2015). *Accuracy of Alcohol Use Disorders Identification Test (AUDIT) for detecting problem drinking in 18–35 year-olds in England*. London: Alcohol Research UK.
- Furnell, S.M., Bryant, P., & Phippen, A.D. (2007). Assessing the security perceptions of personal internet users. *Computers and Security*, 26, 410–417.
- Gordon, R. (2011). An audit of alcohol brand websites. *Drug and Alcohol Review*, 30, 638–644.
- Gordon, R., Harris, F., Mackintosh, A.M., & Moodie, C. (2011). Assessing the cumulative impact of alcohol marketing on young people's drinking: Cross-sectional data findings. *Addiction Research and Theory*, 19, 66–75.
- Gordon, R., Hastings, G., & Moodie, C. (2010a). Alcohol marketing and young people's drinking: What the evidence base suggests for policy. *Journal of Public Affairs*, 10, 88–101.
- Gordon, R., Mackintosh, A.M., & Moodie, C. (2010b). The impact of alcohol marketing on youth drinking behaviour: A two-stage cohort study. *Alcohol and Alcoholism*, 45, 470–480.
- Griffiths, R., & Casswell, S. (2010). Intoxicogenic digital spaces? Youth, social networking sites and alcohol marketing. *Drug and Alcohol Review*, 29, 525–530.
- Grube, J., & Waiters, E. (2005). Alcohol in the media: Content and effects on drinking beliefs and behaviors among youth. *Adolescent Medicine Clinics*, 16, 327–343.
- Hastings, G., Brooks, O., Stead, M., Angus, K., Anker, T., & Farrell, T. (2010). “They'll Drink Bucket Loads of the Stuff”: An analysis of internal alcohol industry advertising documents. London: Alcohol Education Research Council.
- Hastings, G., & Sheron, N. (2013). Alcohol marketing: Grooming the next generation. *British Medical Journal*, 346, f1227.
- Jernigan, D.H. (2009). The global alcohol industry: An overview. *Addiction*, 104, 6–12.
- Jernigan, D.H., & Rushman, A.E. (2014). Measuring youth exposure to alcohol marketing on social networking sites: Challenges and prospects. *Journal of Public Health Policy*, 35, 91–104.
- Jones, S.C., & Magee, C.A. (2011). Exposure to alcohol advertising and alcohol consumption among Australian adolescents. *Alcohol and Alcoholism*, 46, 630–637.
- Lin E.-Y., Caswell, S., You, R.Q., & Huckle, T. (2012). Engagement with alcohol marketing and early brand allegiance. *Addiction Research and Theory*, 20, 329–338.
- Lyons, A.C., McCreanor, T., Hutton, F., Goodwin, I., Barnes, H.M., Griffin, C., ... Samu, L. (2014). *Flaunting it on Facebook: Young adults, drinking cultures and the cult of celebrity*. Wellington: Massey University School of Psychology.
- MacLeod, I. (2013a). “We'll see digital at the heart of our brand activations” reveals SABMiller Europe's head of digital enablement Gavin Marshall. *The Drum*. Retrieved from <http://www.thedrum.com/news/2013/04/09/we-ll-see-digital-heart-our-brand-activations-reveals-sabmiller-europe-s-head>.
- MacLeod, I. (2013b). WKD teams with Spotify for new interactive SoundClash events campaign. *The Drum*. Retrieved from <http://www.thedrum.com/news/2013/10/16/wkd-teams-spotify-new-interactive-soundclash-events-campaign>.
- McClure, A.C., Stoolmiller, M., Tanski, S.E., Engels, R.C.M.E., & Sargent, J. (2013). Alcohol marketing receptivity, marketing-specific cognitions, and underage binge drinking. *Alcoholism: Clinical and Experimental Research*, 37, 404–413.

- McCracken, J.C., & Macklin, M.C. (1998). The role of brand names and visual cues in enhancing memory for consumer packaged goods. *Marketing Letters*, 9, 209–226.
- McCreanor, T., Lyons, A., Griffin, C., Goodwin, I., Barnes, H.M., & Hutton, F. (2013). Youth drinking cultures, social networking and alcohol marketing: Implications for public health. *Critical Public Health*, 23, 110–120.
- Moreno, M.A., Christakis, D.A., Egan, K.G., Brockman, L.N., & Becker, T. (2012). Associations between displayed alcohol references on Facebook and problem drinking among college students. *Achieves Paediatric Adolescent Medicine*, 166, 157–163.
- Morgan, E.M., Snelson, C. & Elison-Bowers, P. (2010). Image and video disclosure of substance use on social media websites. *Computers in Human Behaviour*, 26, 1405–1411.
- Mullman, J. (2009). Southern Comfort pours entire media budget into digital. *Advertising Age*. Retrieved from <http://adage.com/article/news/southern-comfort-pours-entire-media-budget-digital/138202/>.
- National Institute for Health and Care Excellence. (2010). *Alcohol-use disorders: Preventing harmful drinking*. London: National Institute for Health and Care Excellence.
- Nicholls, J. (2012). Everyday, everywhere: Alcohol marketing and social media – current trends. *Alcohol and alcoholism*, 47, 486–493.
- Norman, J., Grace, S., & Lloyd, C. (2014). Legal high groups on the internet: The creation of new organised deviant groups? *Drugs: Education, Prevention and Policy*, 21, 14–23.
- Nudd, T. (2012). Guinness QR Cup reveals scan-able code when full: Another reason to choose the black stuff. *Adweek*. Retrieved from <http://www.adweek.com/adfreak/guinness-qr-cup-reveals-scannable-code-when-full-140602>.
- OFCOM. (2014). *Children and parents: Media use and attitudes report, 2014*. London: OFCOM.
- OFCOM. (2015a). *The communications market 2015*. London: OFCOM.
- OFCOM. (2015b). *Adults' media use and attitudes report, 2015*. London: OFCOM.
- Office for National Statistics. (2015a). *Statistical bulletin: Internet users, 2015*. London: Office for National Statistics.
- Office for National Statistics. (2015b). *Adult drinking habits in Great Britain, 2013*. London: Office for National Statistics.
- Pontes, H.M, Szabo, A., & Griffiths, M.D. (2015). The impact of internet-based specific activities on the perception of internet addiction, quality of life and excessive usage: A cross-sectional study. *Addictive Behaviours Reports*, 1, 19–25.
- Public Health England. (2014). *Alcohol treatment in England, 2013–2014*. London: Public Health England.
- Purves, R.I, Stead, M., & Eadie, D. (2014). *What are you meant to do when you it everywhere? Young people, alcohol packaging and digital media*. London: Alcohol Research UK.
- Rehm, J., & Shield, K.D. (2013). Alcohol and mortality: Global alcohol-attributable deaths from cancer, liver cirrhosis and injury in 2010. *Current Reviews*, 35, 174–183.
- Ridout, B., Campbell, A., & Ellis, L. (2012). 'Off your Face(book)': Alcohol in online social identity construction and its relation to problem drinking in university students. *Drug and Alcohol Review*, 31, 20–26.
- Schultz, E. (2014). MillerCoors turns web into barroom with digital ad deals. *Advertising Age*. Retrieved from <http://adage.com/article/digital/millercoors-strikes-a-series-digital-ad-deals/292191/>.
- Schultz, E. (2015). Bullish on Digital, Heineken USA reveals 2015 ad plans. *Advertising Age*. Retrieved from <http://adage.com/article/cmo-strategy/bullish-digital-heineken-usa-reveals-2015-ad-plans/297444/>.
- Shield, K.D., Parry, C., & Rehm, J. (2013). Chronic diseases and conditions related to alcohol use. *Current Reviews*, 35, 155–173.
- Siegel, M., Kurland, K.P., Castrini, M., Morse, C., de Groot, A., Retamozo, C., . . . Jernigan, D.H. (2015). Potential youth exposure to alcohol advertising on the internet: A study of internet versions of popular television programs. *Journal of Substance Use* [Online]. doi: 10.3109/14659891.2015.1029023.
- Smith, L.A., & Foxcroft, D.R. (2009). The effect of alcohol advertising, marketing and portrayal on drinking behaviour in young people: Systematic review of prospective cohort studies. *BMC Public Health*, 9, 1–11.
- Spencer, A. (2014). Carlsberg beermats get connected with NFC, QR codes and Bluetooth. *Mobile Marketing Magazine*. Retrieved from <http://mobilemarketingmagazine.com/carlsberg-beermats-get-connected-with-nfc-qr-codes-and-bluetooth/>.
- Winpenny, E.M., Marteau, T.M., & Nolte, E. (2013). Exposure of children and adolescents to alcohol marketing and social media websites. *Alcohol and Alcoholism*, 49, 154–159.
- World Health Organisation. (2014). *Global status report on alcohol and health 2014*. Geneva: World Health Organisation.